

Experiment ‘p_4_4_month_with_highest_temperature_array’ Results

June 5, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing assume : 0
Responses containing expect : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_4_4_month_with_highest_temperature_array
Natural language statement: null
Method signature: p_4_4_month_with_highest_temperature_array(inputs: array<real>) returns (hottest: int)

Ensures

- `1 <= hottest <= 12`
- `forall i :: 0 <= i <= inputs.Length - 1 ==> inputs[hottest - 1] >= inputs[i]`

Requires

- `inputs.Length == 12`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_4_4_month_with_highest_temperature_array(inputs: array<real>) returns (hottest: int)`

The method should respect the following contract:

requires `inputs.Length == 12`, ensures `1 <= hottest <= 12`, ensures forall `i :: 0 <= i <=`
 `inputs.Length - 1 ==> inputs[hottest - 1] >= inputs[i]`

Produce and show only the Dafny body of this method, including the curly braces that
 `surround it`. Do not show the signature nor contract. You must not use 'assume' nor
 `'expect'` anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnM1YA1MBz2akYAJ0XZuEyOTcqm0u

Model: gpt-5.5-2026-04-23

Created at: 1780655104

Finish reason: stop

Usage: 326 tokens in, and 6717 tokens out

```
{
  hottest := 1;
  var idx := 1;

  forall i | 0 <= i < idx
    ensures inputs[hottest - 1] >= inputs[i]
  {
    assert i == 0;
    assert hottest - 1 == 0;
    assert inputs[hottest - 1] == inputs[i];
  }

  while idx < inputs.Length
    invariant 1 <= hottest <= inputs.Length
    invariant 1 <= idx <= inputs.Length
    invariant forall i :: 0 <= i < idx ==> inputs[hottest - 1] >= inputs[i]
    decreases inputs.Length - idx
  {
    if inputs[idx] > inputs[hottest - 1] {
      var previousHottest := hottest;
```

```

    assert inputs[idx] > inputs[previousHottest - 1];
    assert forall i :: 0 <= i < idx ==> inputs[previousHottest - 1] >= inputs[i];

    hottest := idx + 1;
    assert hottest - 1 == idx;
    assert inputs[hottest - 1] == inputs[idx];
    assert inputs[hottest - 1] >= inputs[idx];

    forall i | 0 <= i < idx
      ensures inputs[hottest - 1] >= inputs[i]
    {
      assert inputs[previousHottest - 1] >= inputs[i];
      assert inputs[idx] > inputs[previousHottest - 1];
      assert inputs[idx] >= inputs[i];
      assert inputs[hottest - 1] == inputs[idx];
    }
  } else {
    assert !(inputs[idx] > inputs[hottest - 1]);
    assert inputs[idx] <= inputs[hottest - 1];
    assert inputs[hottest - 1] >= inputs[idx];
  }

  assert inputs[hottest - 1] >= inputs[idx];
  assert forall i :: 0 <= i < idx ==> inputs[hottest - 1] >= inputs[i];

  forall i | 0 <= i < idx + 1
    ensures inputs[hottest - 1] >= inputs[i]
  {
    if i < idx {
      assert inputs[hottest - 1] >= inputs[i];
    } else {
      assert i == idx;
      assert inputs[hottest - 1] >= inputs[idx];
    }
  }
}

idx := idx + 1;
}

assert idx == inputs.Length;
assert inputs.Length == 12;
assert 1 <= hottest <= 12;

forall i | 0 <= i <= inputs.Length - 1
  ensures inputs[hottest - 1] >= inputs[i]
{
  assert i < inputs.Length;
  assert i < idx;
  assert inputs[hottest - 1] >= inputs[i];
}
}

```

Final Program

```
method p_4_4_month_with_highest_temperature_array(inputs: array<real>) returns (hottest:
    ↪ int)
    requires inputs.Length == 12
    ensures 1 <= hottest <= 12
    ensures forall i :: 0 <= i <= inputs.Length - 1 ==> inputs[hottest - 1] >= inputs[
        ↪ i]
{
    hottest := 1;
    var idx := 1;

    forall i | 0 <= i < idx
        ensures inputs[hottest - 1] >= inputs[i]
    {
        assert i == 0;
        assert hottest - 1 == 0;
        assert inputs[hottest - 1] == inputs[i];
    }

    while idx < inputs.Length
        invariant 1 <= hottest <= inputs.Length
        invariant 1 <= idx <= inputs.Length
        invariant forall i :: 0 <= i < idx ==> inputs[hottest - 1] >= inputs[i]
        decreases inputs.Length - idx
    {
        if inputs[idx] > inputs[hottest - 1] {
            var previousHottest := hottest;
            assert inputs[idx] > inputs[previousHottest - 1];
            assert forall i :: 0 <= i < idx ==> inputs[previousHottest - 1] >= inputs[i];

            hottest := idx + 1;
            assert hottest - 1 == idx;
            assert inputs[hottest - 1] == inputs[idx];
            assert inputs[hottest - 1] >= inputs[idx];

            forall i | 0 <= i < idx
                ensures inputs[hottest - 1] >= inputs[i]
            {
                assert inputs[previousHottest - 1] >= inputs[i];
                assert inputs[idx] > inputs[previousHottest - 1];
                assert inputs[idx] >= inputs[i];
                assert inputs[hottest - 1] == inputs[idx];
            }
        } else {
            assert !(inputs[idx] > inputs[hottest - 1]);
            assert inputs[idx] <= inputs[hottest - 1];
            assert inputs[hottest - 1] >= inputs[idx];
        }

        assert inputs[hottest - 1] >= inputs[idx];
        assert forall i :: 0 <= i < idx ==> inputs[hottest - 1] >= inputs[i];
    }
}
```

```

forall i | 0 <= i < idx + 1
  ensures inputs[hottest - 1] >= inputs[i]
{
  if i < idx {
    assert inputs[hottest - 1] >= inputs[i];
  } else {
    assert i == idx;
    assert inputs[hottest - 1] >= inputs[idx];
  }
}

idx := idx + 1;
}

assert idx == inputs.Length;
assert inputs.Length == 12;
assert 1 <= hottest <= 12;

forall i | 0 <= i <= inputs.Length - 1
  ensures inputs[hottest - 1] >= inputs[i]
{
  assert i < inputs.Length;
  assert i < idx;
  assert inputs[hottest - 1] >= inputs[i];
}
}

```

Total Token Usage

Input tokens: 326

Output tokens: 6717

Reasoning tokens: 5987

Sum of 'total tokens': 7043

Experiment Timings

Overall Experiment started at 1780655102802, ended at 1780655277410, lasting 174608ms (174.61 seconds)

Iteration #1 started at 1780655102805, ended at 1780655277410, lasting 174605ms (174.61 seconds)